PALEOMAGNETIC STUDY OF SEDIMENTS AROUND THE KOMYOIKE VOLCANIC ASH HORIZON IN OSAKA GROUP, SENPOKU AREA, OSAKA PREFECTURE, JAPAN

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Abstract. The Osaka Group is typical of the Plio-Pleistocene Series in Japan. The correlation of paleomagnetic stratigraphy and biostratigraphy gives a clear picture of the Quaternary history of the Group. The Komyoike volcanic ash layer distributed only in the Senpoku area, in the southern part of Osaka Prefecture is shown to have normal polarity at 1.1 million years (fission track age). The remanent magnetizations of more than one hundred oriented samples collected in sequence from the out-crop sediments intercalated with the Komyoike volcanic ash bed were measured. Paleomagnetic results after magnetic cleaning show that the sedimentary sequence is divided into three distinct zones by changes in the remanent polarity.

Introduction

A clear picture of the Quaternary history has been reconstructed for the Plio-Pleistocene Series widely distributed in south-western Japan. One of them, the Osaka Group is composed of unconsolidated gravels, sands, silt and clay beds of limnic, fluvial and marine origin with many volcanic ash layers. A magnetostratigraphic approach has contributed to the dating of the fossil vertebrates and the fossil plants described from the Group [Maenaka et al., 1977]. The upper half of the Osaka Group consists of alternating bands of marine and nonmarine sediments. Most of the marine facies are represented by continuous clay marker beds differentiated by enclosed pyroclastic layers specified as Ma 0 through Ma 13 in ascending order. It was suggested that the global glacio-eustatic mechanism caused the oscillations of marine and nonmarine deposition within the Osaka Group. This occurred at least ten times between approximately 1.5 and 0.27 million years. The above description is based on the paleomagnetic stratigraphy presented by the writer with his collaborators (Fig. 1). As shown in Fig. 1, the paleomagnetic polarity of the Komyoike volcanic ash layer in the Osaka Group is normal, and the fission track age is 1.1 million years old. This age given by Nishimura and Sasajima [1970] is considered to be a reasonable one based on geological grounds. The writer suggested that a paleomagnetic study on the lacustrine and marine sediments accumulated at high sedimentation rates, giving a continuous record of the geomagnetic directional change, might have revealed an unknown event in the Matuyama Epoch [Maenaka, 1975].

Geological Outline

The Plio-Pleistocene Osaka Group distributes in the central Kinki district, Japan. The stratigraphic sequence of the Osaka Group has been worked out in detail. The Osaka Group is composed of unconsolidated sediments such as gravels, sands, silt, clay and volcanic ash. The volcanic ash layers are the most useful key beds as they possess individual characteristics such as coloration, relative thickness, grain size, internal stratification, heavy mineral composition, general shape and refractive index of glass shards, and the thermomagnetic curves of ferromagnetic minerals. The age of deposition of individual volcanic ash layers is obtained by the dating method of fission track and paleomagnetic chronology.

Palecmagnetic data of volcanic ashes dated by fission track is shown in Fig. 1. Considering the paleomagnetic data and fission track age, the layers above the Fukakusa volcanic ash are correlated with the Brunhes Normal Epoch. Black silt in Ma 4 below the Fukakusa volcanic ash layer has reversed polarity. Consequently, the Brunhes-Matuyama boundary in the Osaka Group is considered to lie between the Fukakusa volcanic ash layer and the Ma 4 marine clay bed.

The layers between the Azuki and Shimakumayama ash beds in the Osaka Group is correlated with the Matuyama Reversed Epoch. The normal polarity events of the Matuyama Epoch are represented by normally magnetized volcanic ashes such as Kamikatsura and Komyoike-Pink ashes. The Kamikatsura ash is situated a few meter above Ma 2. The fission track age of the Azuki ash, intercalated in Ma 3, is 0.87 my. Therefore the age of Kamikatsura ash may be about 0.95 my. The normal polarity of the Kamikatsura ash is correlated with the Jaramillo Normal Event. The age of the Komyoike volcanic ash is 1.1 my. This age is considered to be a reasonable age, judging from the thickness of the sediments and the age of the intercalated volcanic ashes; Kasuri (0.38 my fission track age), Fukakusa (0.69 my paleomagnetic age), Azuki (0.87 my fission track age), Grey (1.5 my fission track age), and Shimakumayama (2.3 my fission track age). This means that a new event may be present at 1.1 million year.

The Komyoike volcanic ash layer is imbedded between the Pink volcanic ash bed and Ma 2 bed, and intermittently distributed only in the Senpoku area in the southern part of Osaka Prefecture. The Pink volcanic ash bed is two to five meters above Ma 1 bed, and is widely distributed. The extracted ferromagnetic minerals from the Pink volcanic ash are characterized by the presence of ilmenite [Maenaka and Yokoyama, 1972].

Sampling and Measurement

Sampling was carried out on out-crop in the east of Mikita, Izumi City in the Senpoku area. Twenty oriented samples were taken from the Pink horizon sediments of six meter length. From the Komyoike horizon sediments of twenty meter length,

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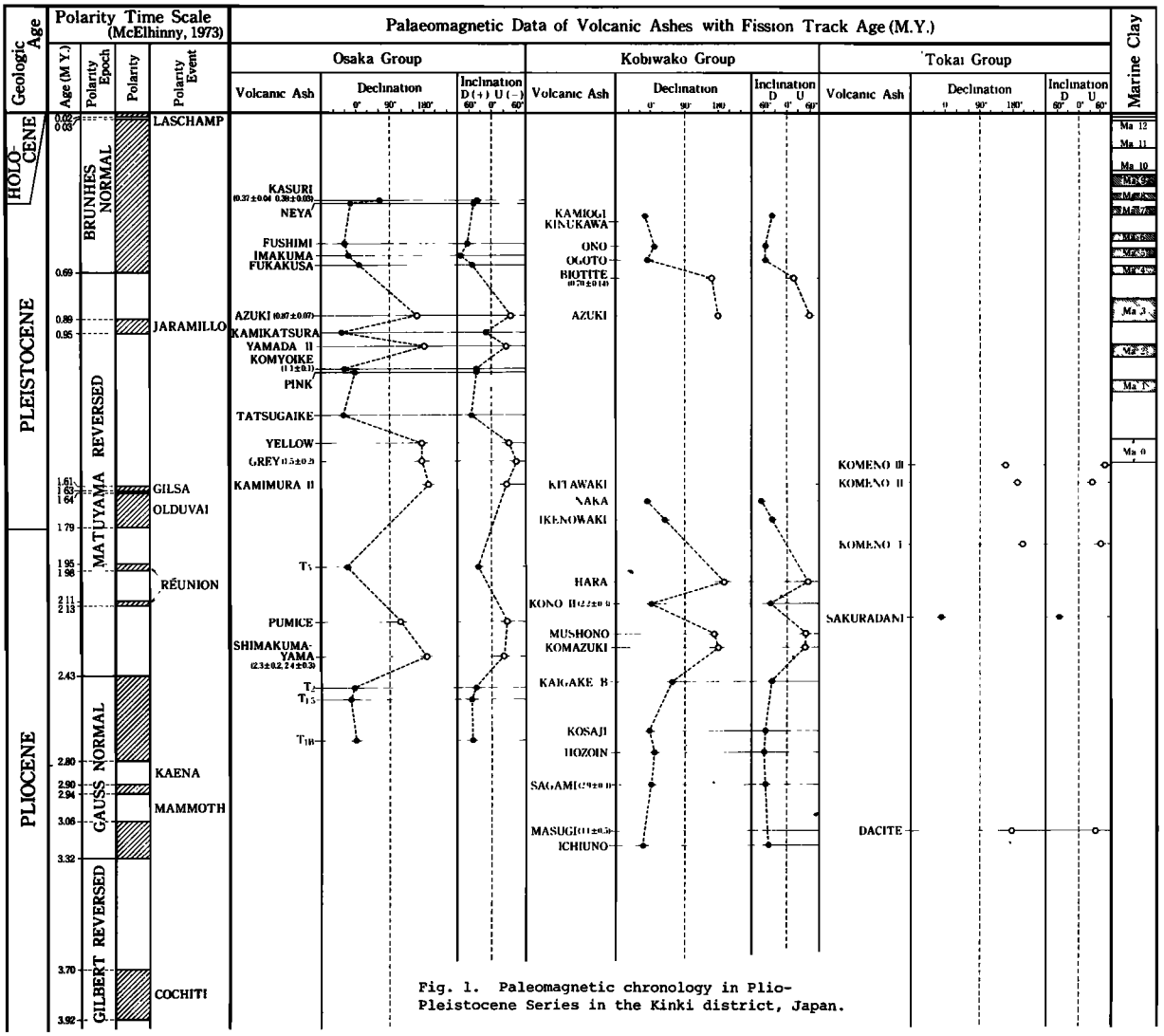


Fig. 1. Paleomagnetic chronology in Plio-Pleistocene Series in the Kinki district, Japan.

more than one hundred oriented samples were collected successively. The stratigraphic positions of the collected samples are illustrated in Figs. 2 and 3. From each sample, several adjacent cubes one inch on a side were prepared for the purpose of the paleomagnetic study. The directions of the remanent magnetizations of the prepared specimens were measured by a spinner magnetometer at Kyoto University. By using a spinner magnetometer, the measurement of the weak remanent magnetization of the Osaka Group sediments whose intensity is less than 1×10^{-7}emu/g became possible. From the result of the preliminary test using a pilot specimen, magnetic cleaning of all specimens were done in an alternating peak field of 100 oersteds using an instrument equipped with a three axis tumbler.

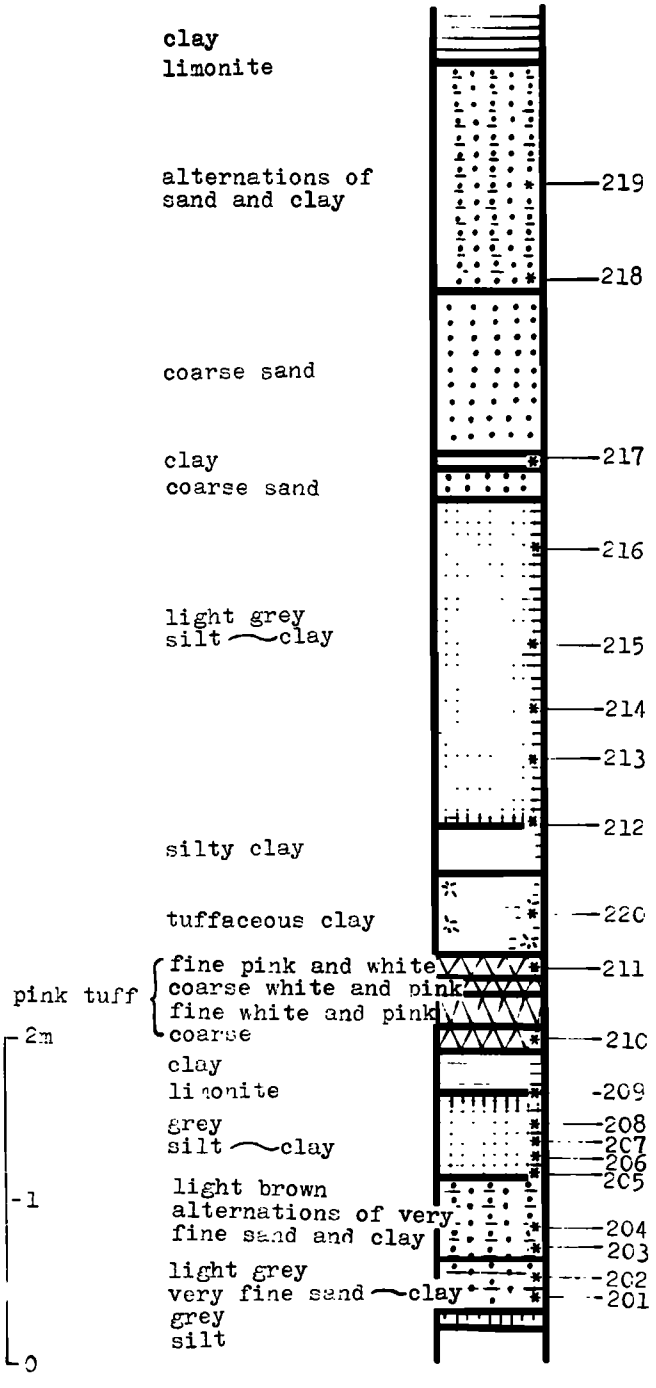


Fig. 2. Stratigraphic column of the Pink horizon sediments.

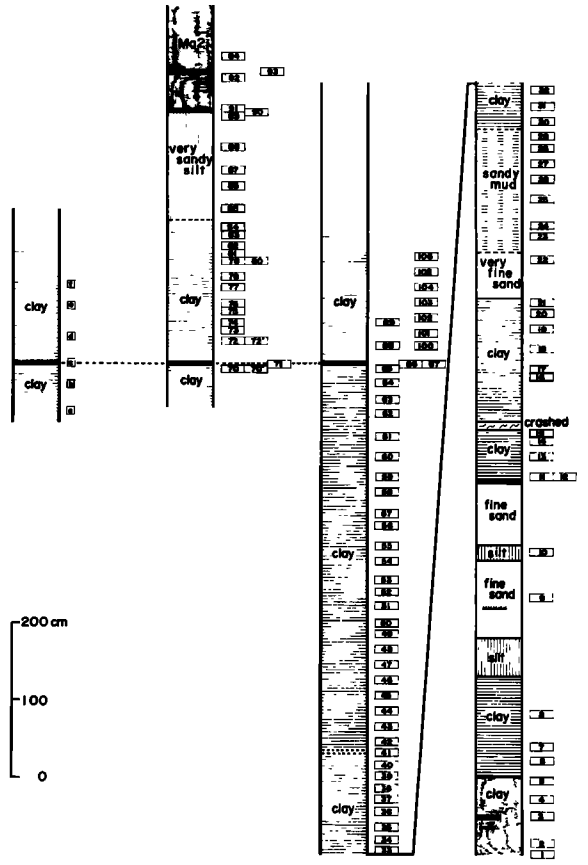


Fig. 3. Stratigraphic column of the Komyoike horizon sediments.

Discussion

The value of the mean direction of the remanent magnetization of individual samples after

Table 1. Paleomagnetic data after magnetic cleaning for Pink horizon sediments.

Sample Number	Number of Specimens	Mean Direction Decli.	Incli.	k	VPP Lat.	Long.
219	10	19.7E	68.8	19	67.4N	168.0E
218	6	29.7E	40.2	18	61.6N	119.3W
217	10	21.7W	78.2	4	54.9N	121.1E
216	7	13.9E	54.2	25	78.5N	128.5W
215	5	22.9E	48.4	10	69.9N	125.9W
214	4	0.4W	63.5	17	79.4N	134.0E
213	4	27.6E	23.7	93	56.6N	99.8W
212	3	2.5W	36.9	18	75.9N	34.8W
209	6	13.9E	45.0	42	75.7N	104.7W
208	5	11.7E	52.0	36	74.3N	88.3W
207	6	8.8E	47.3	72	80.4N	98.3W
206	5	14.1E	45.7	12	75.8N	101.7W
205	6	3.8W	39.1	10	77.2N	27.8W
204	5	6.3E	44.6	35	80.1N	79.4W
203	8	29.5E	25.4	24	56.0N	103.4W
202	8	4.2W	42.8	8	79.6N	22.8W
201	4	2.5W	43.3	59	80.4N	30.7W

magnetic cleaning in A.C. field of 100 oersteds, the precission parameter k, and the virtual pole position are given in Tables 1 and 2.[1] The sample numbers of the first column in these tables are arranged in descending order as shown in Figs. 2 and 3. The samples whose remanent magnetizations were not measured, for example due to crushing of the specimens, are omitted from the tables (210, 211 and 220 in Fig. 2, 1, 65, 68, 87, 89, 106, b, and d in Fig. 3). The star in the last column in Table 2 shows that the data is probably not suggestive of any mean direction due to its great scatter.

All samples of the Pink horizon sediments have the remanent magnetization directed to the present geomagnetic field, and the grouping of the direction of the remanent magnetization of the individual samples is rather good compared to that of the Komyoike horizon sediments. The grouping of the direction of the remanent magnetization of the individual samples collected from the Komyoike horizon sediments is not as good, and the direction of the remanent magnetization is highly variable. However, the stratigraphical sequence of the Komyoike horizon sediments is roughly divided into three divisions by the polarity changes; the normal polarity zone, the transitional polarity zone, and the reversed polarity zone in ascending order. The normal polarity of the lower stratigraphic position below Sample Number 25 continues to the normal polarity of the Pink horizon sediments. The long sequence of normal polarity ranging from the Pink ash to the Komyoike ash was as expected. The writer tentatively names this long sequence of normal polarity dated about 1.1 million years old as "Komyoike event" after the type locality of the Komyoike volcanic ash layer. The reversed polarity of the higher ones above Sample Number 48 is thought to continue to the reversed polarity of the Yamada volcanic ash layer in Ma 2 bed. The middle part between Sample Number 26 and 47 is considered to be a transitional period from a stable normal field to a stable reversed field.

The value of the virtual pole position marked by the star in Table 2 is, roughly speaking, in low latitude. It is suggested that the variations are due to the nondipole field when the reduction of the dipole field strength during the field reversal caused the dispersion of the remanent magnetization of the sediments.

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[1] Table 2 is available on microfiche. Order from American Geophysical Union, 1909 K Street, N.W., Washington, D.C. 20006. Document L79-004; $1.00. Payment must accompany order.